

Ecosystems

Starter:

1. What is an abiotic factor?

A non-living part

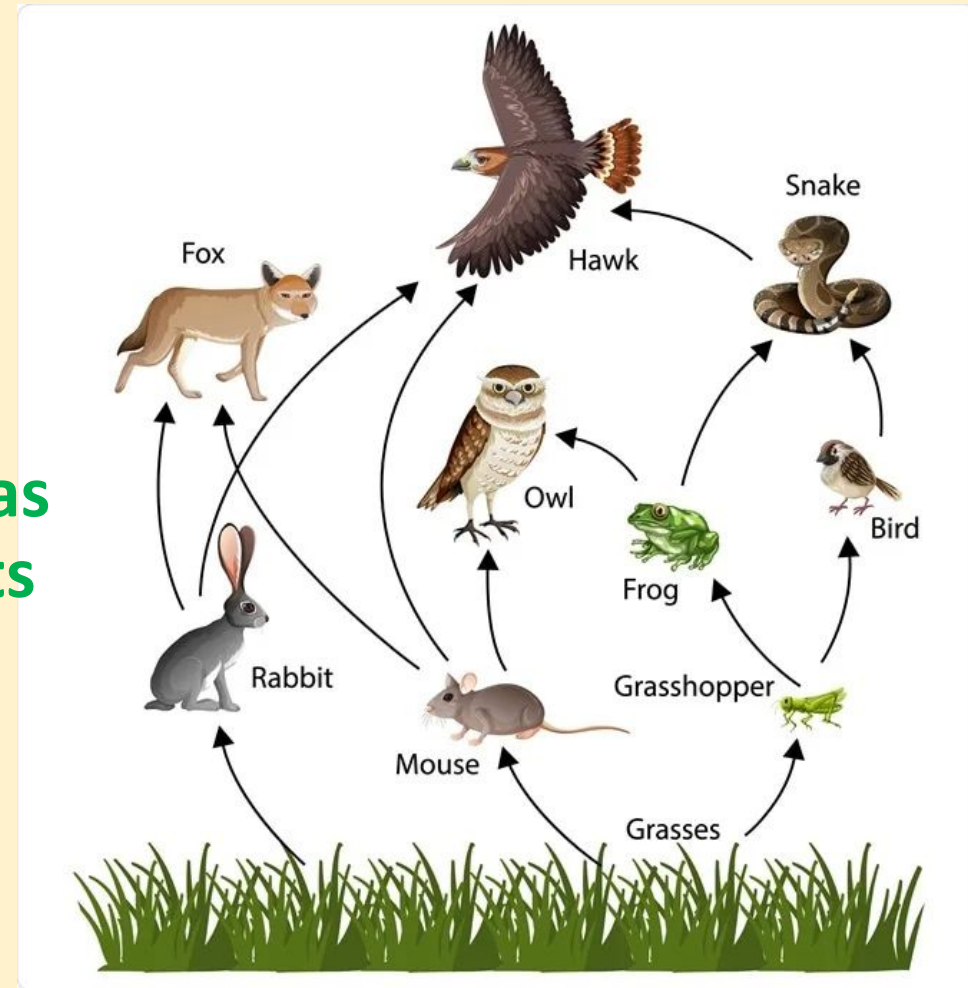
1. What is the difference between a habitat and an ecosystem?

A habitat refers to where an organism lives whereas an ecosystem includes all living and non-living parts

1. How many secondary consumers are there in this food web?

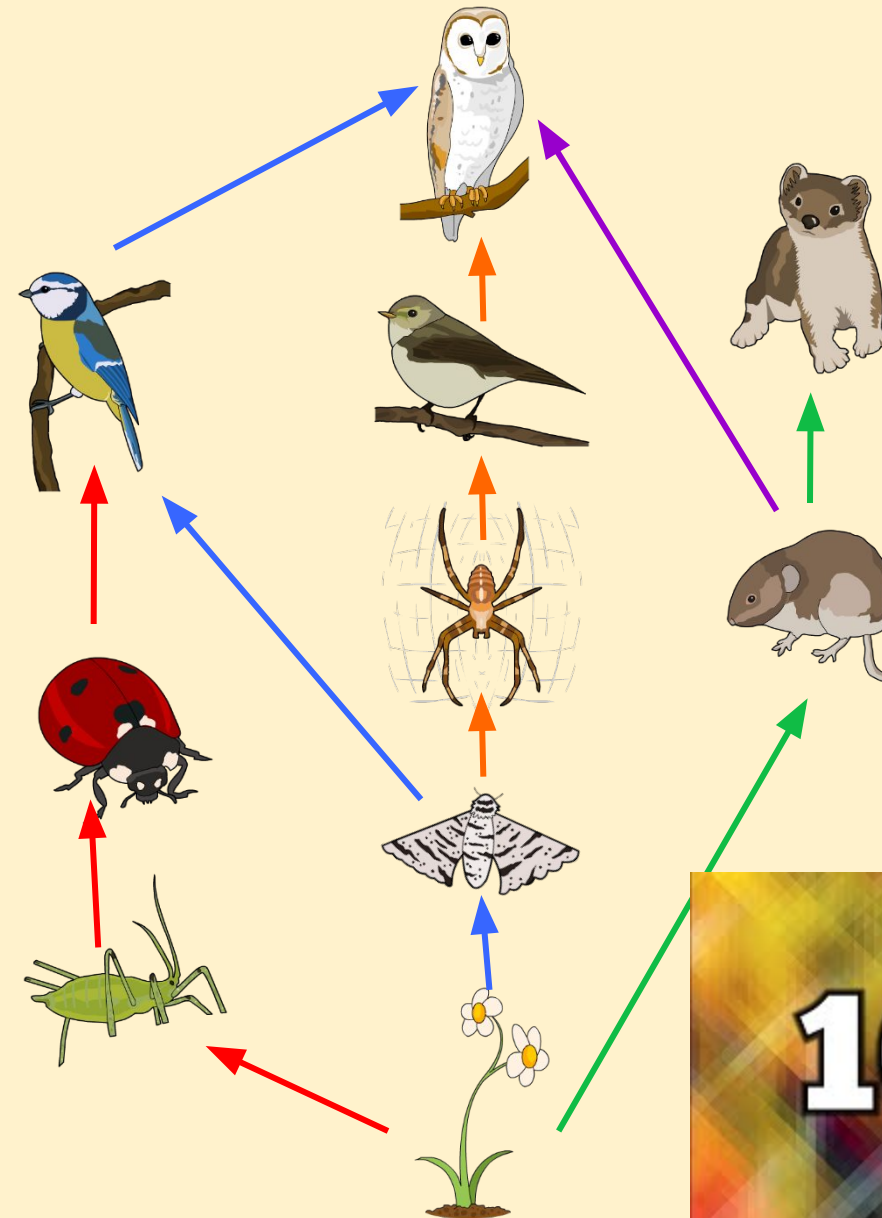
4 (fox, owl, frog, bird)

1. Draw / write a food chain that involves the frog



Food webs

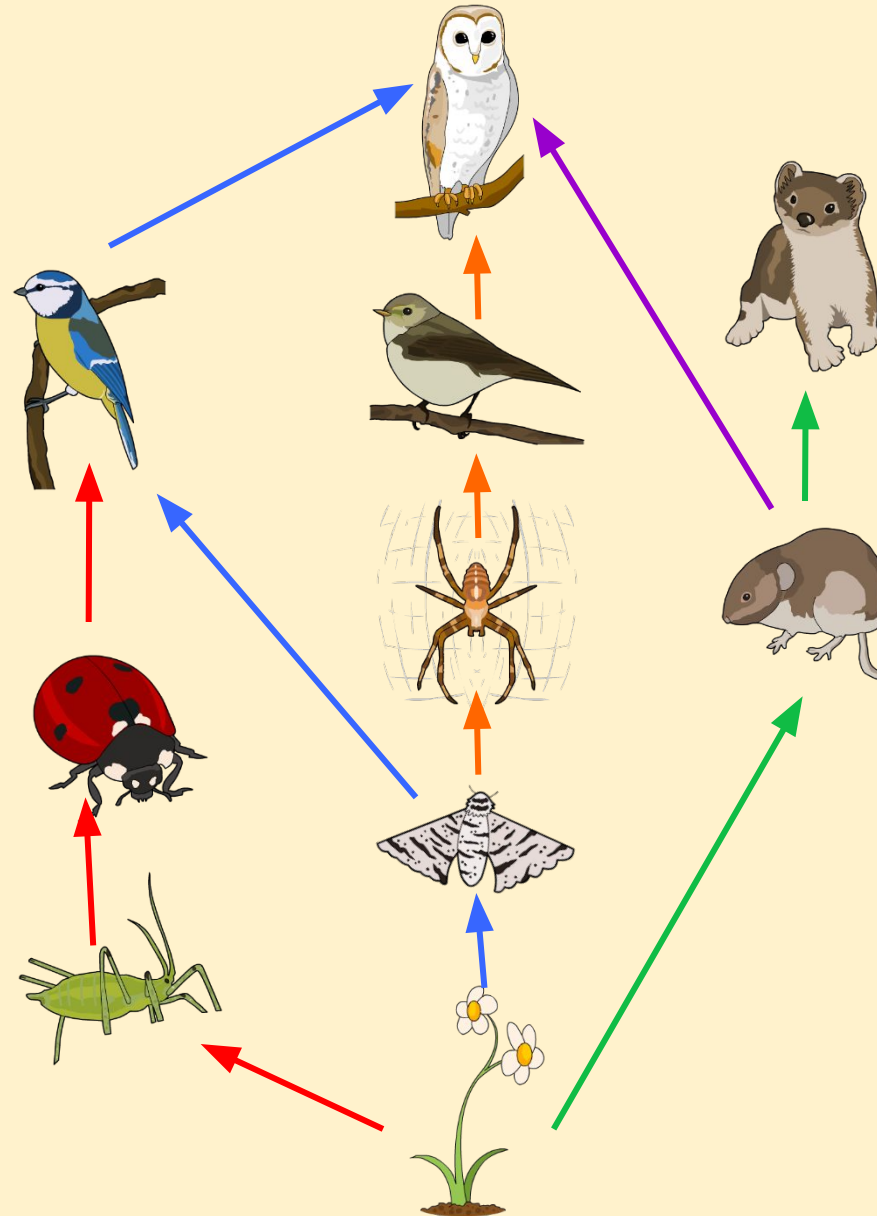
1. Name the producer in this food web.
2. Name two herbivores in this food web.
3. Name two species that are tertiary consumers
4. How many secondary consumers are there?
5. What would happen if most blue tits disappeared because of a disease?



10:00

Food webs

1. Name the producer in this food web. **Daisy(flower/plant)**
2. Name two herbivores in this food web. **Vole and Aphid**
3. Name two species that are tertiary consumers. **Goldfinch and barn owl**
4. How many secondary consumers are there? **Four (ladybird, spider, stoat, blue tit)**
5. What would happen if most blue tits disappeared because of a disease? **Ladybug and moth will increase**



Success criteria

Lesson Title	Working towards	Working at	Working beyond
What is an Ecosystem	I can name some living (biotic) and non-living (abiotic) parts of an ecosystem.	I can define key terms like community, population, habitat and explain how energy moves through a food web	I can explain the role of decomposers and interpret pyramids of numbers to show energy transfer in ecosystems.
Interactions within Ecosystems	I can identify some habitats found in the Sonoran Desert and other ecosystems.	I can describe interactions between organisms and their environment in different ecosystems.	I can explain the differences between habitats and ecosystems and how organisms are adapted to survive in these settings.
Adaptations and Competition	I can give examples of how plants or animals are adapted to their environments.	I can describe how adaptations help species survive and explain competition between organisms.	I can analyse mutualistic relationships and present research on how different species interact in ecosystems.
Invasive Species	I can name an invasive species and describe what a native species is.	I can describe how invasive species affect native plants and animals in an ecosystem.	I can explain why controlling invasive species is important and evaluate different methods of control.
Case Study - New Zealand (Debate)	I can identify an invasive species problem in New Zealand.	I can describe the ecological impact of invasive species and outline arguments for control.	I can construct and defend an argument in a debate using evidence about species control and conservation.
Bioaccumulation	I can state what an insecticide is and give a simple example.	I can describe how chemicals like DDT build up in organisms and impact the food chain.	I can explain why organisms at the top of the food chain have higher toxin levels and the environmental consequences.
Weather and Climate	I can describe the difference between weather and climate.	I can explain how weather affects plants and animals in an ecosystem.	I can analyse weather data to predict effects on local ecosystems and seasonal changes.
Climate and Atmosphere	I can name key gases in the atmosphere.	I can explain how greenhouse gases affect Earth's climate and describe human impacts.	I can evaluate how climate change may affect ecosystems and propose possible solutions.

PEEK
3
600



ECOSYSTEM

Note for teachers to make lucidspark document for each class if you want to use it. We do not have jamboard anymore so lucidspark is the next best thing

Research Task: Examples of Ecosystems

What examples of ecosystems can you think of?

For the next task, open Lucidspark (posted on GC)

Link (for 8J)

https://lucid.app/lucidspark/daf1c53c-c5de-4bf2-b60d-d4ebf7028519/edit?invitationId=inv_0e1e494a-a540-491a-b830-7f59e9d6dc7b

Research Task: Examples of Ecosystems

With your partner research information about two of the ecosystems, make notes on Lucidspark. After the time is up you will look at each others notes

1. Mangrove Forest
2. Sea ice in the Arctic Ocean
3. Rice Paddy
4. River

In your notes on Lucidspark, include:

- The different **habitats** within the ecosystem
- What **organisms** live in the habitats (**biotic**)

Stretch: how these organisms **interact** with each other or the ecosystem

- What **abiotic factors** can be found in this ecosystem



Examples of Ecosystems

Read the information on the other ecosystems from lucidspark and write 2-3 sentences in your books to summarise the key information of each ecosystem

Eg.

In a mangrove forest the main habitats are...

Some of the organisms that live in those habitats are...

The way these organisms interact is...



Examples of Ecosystems

Read the information on the other ecosystems from the jamboard and write 2-3 sentences in your books to summarise the key information of each ecosystem

Eg.

In a mangrove forest the main habitats are... ponds, lakes, rivers

Some of the organisms that live in those habitats are... beavers, tigers, crabs, crocodiles

The way these organisms interact is... that the predators such as crocodiles will eat the prey animals such as beavers and crabs

Let's practice

Answer the questions in your workbook on ecosystems.

Exercise 4.2

Workbook pg 72 - 75



> 4.2 Different ecosystems

Exercise 4.2A A tropical rainforest ecosystem

Focus

This exercise gives you practice in using the correct words when you are writing about ecosystems and habitats.

Complete the sentences. Use each of these words once only.

decomposers ecosystem environment habitats nutrients

Tropical rainforests grow where the temperature is always high and there is plenty of rainfall. Many different species of plant grow in the rainforest.

The rainforest provides and food for many different species of animal.

On the forest floor, fungi break down dead leaves and waste from the animals. These fungi are

The fungi release from the dead leaves and waste, which help the plants to grow.

All of the plants, animals and fungi interact with one another. They also interact with the non-living parts of their

This network of interactions makes up the tropical rainforest

Exercise 4.2B Hydrothermal vents

Practice

In this exercise, you will think about a very unusual ecosystem. You will use information to construct a food chain, and think about the different habitats in the ecosystem.

In 1977, a completely new ecosystem was discovered. In the deep ocean, below the surface, scientists found places where hot water rushes out from the sea bed into the water.

Topic 4.2 Different ecosystems

Exercise 4.2A A tropical rainforest ecosystem

Tropical rainforests grow where the temperature is always high and there is plenty of rainfall. Many different species of plant grow in the rainforest.

The rainforest provides **habitats** and food for many different species of animal.

On the forest floor, fungi break down dead leaves and waste from the animals. These fungi are **decomposers**.

The fungi release **nutrients** from the dead leaves and waste, which help the plants to grow.

All of the plants, animals and fungi interact with one another. They also interact with the non-living parts of their **environment**.

This network of interactions makes up the tropical rainforest **ecosystem**.

Exercise 4.2B Hydrothermal vents

- 1 There is no light, so they cannot photosynthesise.
- 2 For example: bacteria → zooplankton → sea anemones → crabs
- 3 bacteria
- 4 From chemicals dissolved in the hot water that comes out of the vent.
- 5 For example: inside a tube worm; in the hot water around the vent; on the sea floor.

Mark your work using a green pen!

Exercise 4.2C Mangroves and fish

Mark your work using a green pen!

- 1 habitat – the place where an organism lives
ecosystem – a network of interactions between living and non-living things
predator – an animal that kills and eats other animals
- 2 yellowtail
- 3 The masses for all species are lower where there are no mangroves. The biggest difference is for yellowtail, where the mass is less than half of that where there are mangroves on the shore.
- 4 When no mangroves are present, there is no habitat for the baby fish to develop safely from predators, so fewer baby fish grow to become adults, and there are fewer adult fish on the coral reef.